

Meridian Twin

An AI digital twin that shows what a storm, a delay wave or a fuel spike will cost an airline, and the best way out.

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THE PROBLEM: AIRLINES LOSE MONEY IN THREE PLACES

\$347k

Storms
Cost of waiting out an average ground stop day

\$418k

Routine delays
Every day, about 59 late flights at \$98.41 a minute

\$300k

Fuel
Lost per day at the October 6 jet price of \$4.34 a gallon, fares unchanged

Figures are for the 61 aircraft synthetic airline in the prototype.

WHAT IT DOES

- ### 1 Predict

Scores every departure for late risk two hours ahead, using a model trained on real flights.
- ### 2 Simulate

Tests thousands of recovery plans in under a second and shows the cost of each.
- ### 3 Explain and approve

Claude explains the best plan and drafts passenger notices. A person approves every plan.

THE RESULTS

\$97k

saved per storm day, 28% less than waiting it out

36 of 36

simulated storms where the twin's plan beat waiting

77%

late rate for the riskiest tenth of flights, against 28% overall

\$1.2M

saved a year, base case, range \$0.3M to \$2.9M

Delay model tested on 612,038 real July 2026 flights it never saw in training. AUC 0.767.

YEARLY VALUE, STORM RECOVERY ONLY

Airline size	Base case	Range
61 aircraft (the prototype)	\$1.2M	\$0.3M to \$2.9M
150 aircraft	\$2.9M	\$0.7M to \$7.1M
300 aircraft	\$5.7M	\$1.4M to \$14.3M
800 aircraft	\$15.2M	\$3.8M to \$38.0M

Base case: 30 storm days a year and 40% of the gap captured. Low: 15 days, 20%. High: 50 days, 60%. Delays and fuel are not counted.

HOW IT IS BUILT

- Simulator in JavaScript with 8 automated tests
- Delay model in Python on 6.3 million real flights, with weather
- Fuel layer on US EIA spot prices for October 6, 2026
- Claude as the advisor, with a person always approving
- Code, data pipeline and model card are public on GitHub

Why it matters for clients. The airline is the example. The same pattern of predict, simulate, explain and approve carries to energy, healthcare, government and telecom. A first step is one hub for 26 weeks, ending in a saving finance can sign off. These are estimates from a simulated airline, and the pilot replaces them with a measured number.



Try the live product

Scan to open the short presentation with the working twin inside it.

bypxr.github.io/meridian-twin



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